DSAA3073: Theories in Data Science

Week 4 Validation Sheet · From Uniform Guarantees to Model Selection

Coverage: Week 4 **Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. A class has best-in-class population risk 0.18. Which is an agnostic PAC success statement for $(\varepsilon, \delta) = (0.04, 0.05)$?

(a) $R(A(S)) \le 0.04$ for every sample.
(b) $R(A(S)) \le 0.22$ with probability at least 0.95.
(c) $\hat{R}_{S(A(S))} \le 0.22$ with probability at least 0.95.
(d) $\mathbb{E}[R(A(S))] \le 0.22$.

A2 [5 marks]. Why can the uniform event $\sup_{h\in\mathcal{H}}\left|\hat{R}_{S(h)} - R(h)\right| \le \alpha$ be applied to an ERM output h_S selected after seeing S?

(a) ERM is independent of S.
(b) The event holds simultaneously for every $h \in \mathcal{H}$ on the same sample.
(c) ERM always has zero empirical risk.
(d) A pointwise bound for each fixed h automatically gives the same event.

A3 [5 marks]. In which frame does the Week 4 Fundamental Theorem identify finite VC dimension with learnability?

(a) Any prediction problem and any loss.
(b) Distribution-free binary classification with zero-one loss, i.i.d. samples, and suitable measurability.
(c) Only finite hypothesis classes with deterministic algorithms.
(d) Regression with squared loss and arbitrary dependent samples.

A4 [5 marks]. Which description of a ghost sample S' is correct?

(a) It is an independent same-distribution sample introduced only in the proof.
(b) It is a validation set used by the learner to choose a model.
(c) It is a second training set that the algorithm must receive.
(d) It is obtained by relabeling the observed training inputs.

Part B: Definition in use

B1 [20 marks]. Let h_S be an η-approximate ERM in $\mathcal{H}$, so $\hat{R}_{S(h_S)} \le \inf_{h\in\mathcal{H}} \hat{R}_{S(h)} + \eta$. On one sample, the event $\sup_{h\in\mathcal{H}} \left|\hat{R}_{S(h)} - R(h)\right| \le 0.025$ holds and $\eta = 0.008$.

Starting from the definitions, derive the best-in-class population excess-risk certificate for h_S. State the numerical excess bound, explain why the same sample event protects both hypotheses used in your comparison, and handle the possibility that the population infimum is not attained.

Answer space

Part C: Basic skills

C1 [7 marks]. On a fixed sample of size $m = 2$, a loss class has restriction vectors $\mathcal{V} = \{(0,0), (1,0), (0,1)\}$. Compute

$$\mathbb{E}_\sigma\left[\max_{v\in\mathcal{V}} \frac{\sigma_1 v_1 + \sigma_2 v_2}{2}\right]$$

by considering all four equally likely sign vectors.

Answer space

C2 [8 marks]. Three active SRM classes use weights $(w_1, w_2, w_3) = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ and total failure budget $\delta = 0.06$. Their empirical risks are $(0.14, 0.09, 0.04)$ and their valid penalties at the allocated confidences are $(0.04, 0.07, 0.14)$.

Compute the three δ_k, the three certified objectives, and the selected class.

Answer space

Part D: Repair and construction

D1 [17 marks]. A report claims:

"For any prediction problem and any loss, a class is learnable exactly when its VC dimension is finite; moreover one particular ERM rule must succeed."

Rewrite this as a correct Week 4 Fundamental-Theorem statement. Your repair must give the full scope frame and the correct ERM quantifier, and must explain why the original statement cannot be used outside that frame.

Answer space

D2 [18 marks]. A proposed countable SRM scheme activates classes $\mathcal{H}_1, \mathcal{H}_2, \ldots$ and sets $\delta_k = w_k \delta$ with

$$w_k = \frac{3}{2^{k+1}}, \quad k = 1, 2, \ldots.$$

It then advertises simultaneous confidence $1 - \delta$ and gives an active extra class weight zero.

Diagnose both defects. Compute the actual union-bound budget, then construct a valid replacement allocation for all countably many active classes and state how a zero-weight class must be treated.

Answer space

Part E: Proof and derivation

E1 [25 marks]. A measurable binary zero-one loss class is studied on one i.i.d. sample. On an event that holds simultaneously for every $h \in \mathcal{H}$, the available one-sided certificates are

$$R(h) \le \hat{R}_{S(h)} + u(h) \quad \text{and} \quad \hat{R}_{S(h)} \le R(h) + \ell(h),$$

where $u(h), \ell(h) \ge 0$ need not be equal. The learner returns an η-approximate empirical minimizer $\hat{h}$, and for every $\rho > 0$ there is $h_\rho \in \mathcal{H}$ with $R(h_\rho) \le \inf_{h \in \mathcal{H}} R(h) + \rho$.

At the two relevant hypotheses the radii are

$$u(\hat{h}) = 0.040, \quad \ell(\hat{h}) = 0.010, \quad u(h_\rho) = 0.020, \quad \ell(h_\rho) = 0.030,$$

and $\eta = 0.010$. Determine the strongest best-in-class excess-risk certificate justified by this information. Your derivation must make clear which deviation direction is needed at each hypothesis, why simultaneity is essential, and how the unattained-infimum case is handled.

Answer space

Part F: Synthesis and limitations

F1 [15 marks]. Before collecting data, a team declares a fixed library $\mathcal{H}_0$ of 500 binary classifiers and an infinite binary class $\mathcal{H}_\infty$ with finite VC dimension d. Both use zero-one loss and satisfy the Week 4 measurability conditions. From one i.i.d. sample S, the team may select its final classifier from $\mathcal{H}_0 \cup \mathcal{H}_\infty$.

The team observes only 12 distinct loss vectors from $\mathcal{H}_\infty$ on S. It therefore proposes to apply fixed-classifier concentration followed by a union bound over $500 + 12$ observed vectors, claiming a simultaneous population-risk certificate for the final data-selected classifier.

Diagnose this proposal and replace it with a sound Week 4 proof strategy. State precisely which part of the finite-union argument remains valid, why the count 12 does not certify the infinite-class search, and how the final selection can instead receive one simultaneous high-probability guarantee.

Answer space